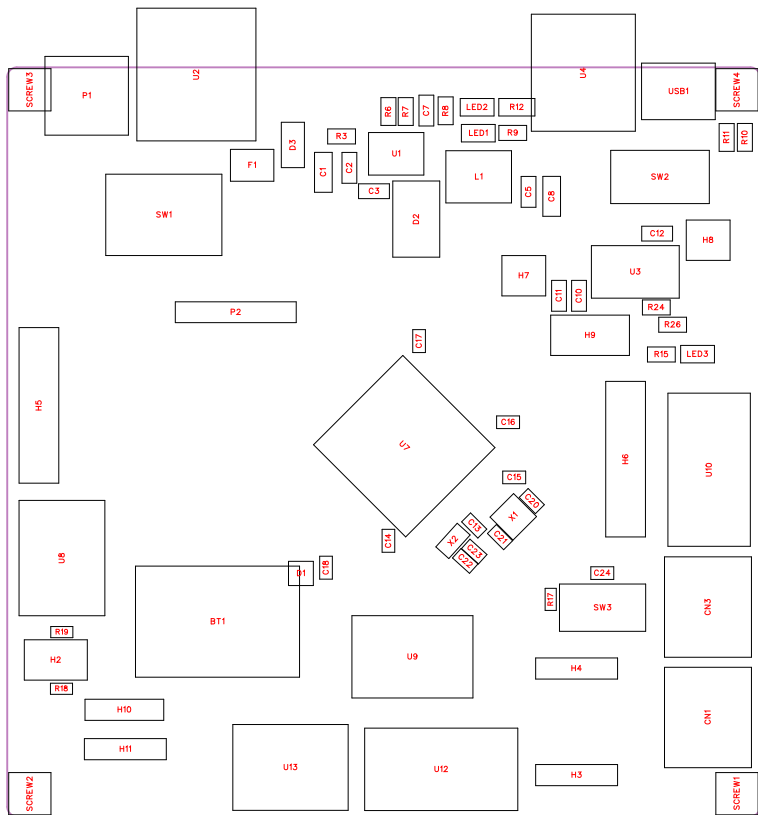
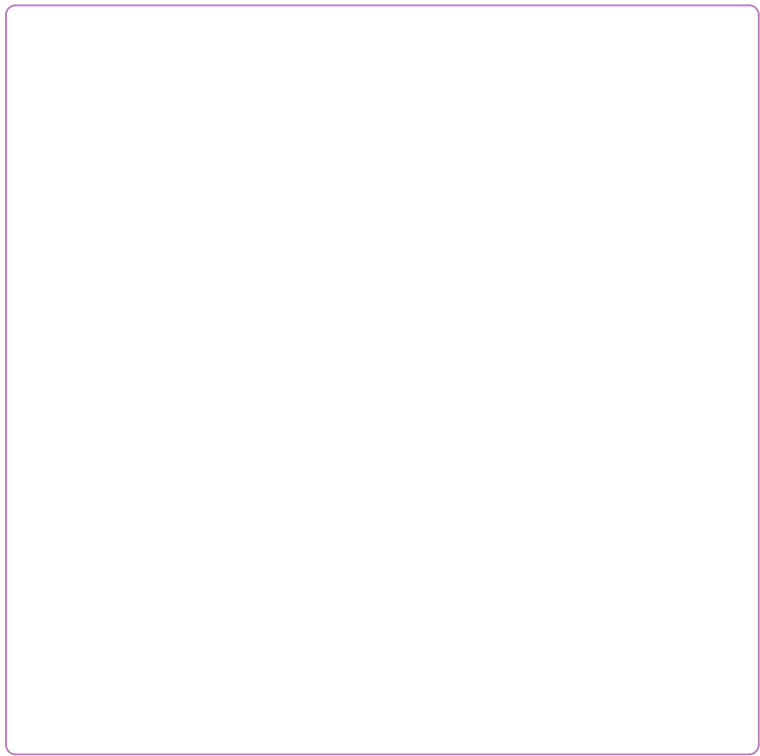


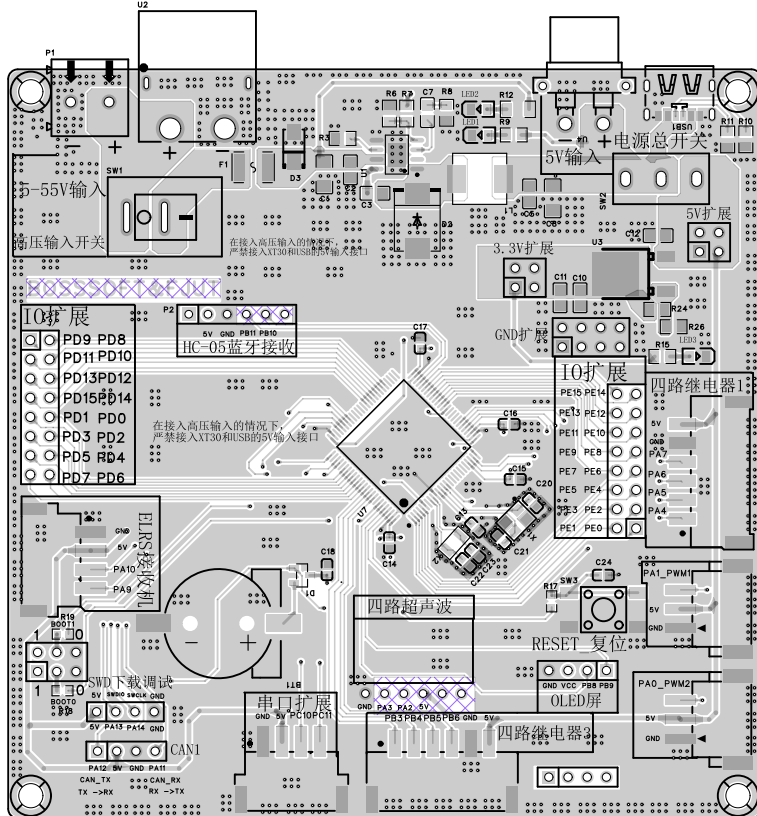


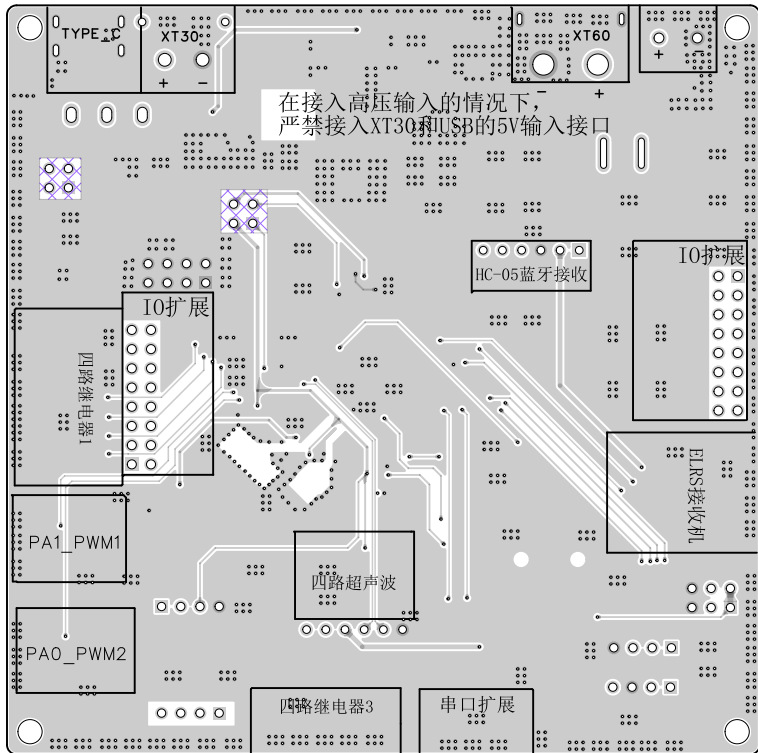


No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier
1	1	CR1220-2ZX	BT1	BAT-SMD_C R1220-2ZX		CR1220-2ZX	Q&J	C969906	LCSC
2	1	4.7uF	C1	C1206	4.7uF	GRM31CC72 A475KE11L	muRata(村田)	C237304	LCSC
3	5	100nF	C2,C3,C5,C10,C12	C0805	100nF	CL21B104K CFNNNE	SAMSUNG(三星)	C28233	LCSC
4	1	330pF	C7	C0805	330pF	CC0805JRN PO9BN331	YAGEO(国巨)	C107116	LCSC
5	1	33uF	C8	C1206	33uF	C3216X5R1 E336MT000 E	TDK	C342307	LCSC
6	1	1uF	C11	C0805	1uF	0805B105K5 00NT	FH(风华)	C215803	LCSC
7	7	100nF	C13,C14,C15,C16,C17,C18 ,C24	C0603	100nF				
8	2	20pF	C20,C21	C0603	20pF				
9	2	10pF	C22,C23	C0603	10pF				
10	2	ZX-XH2.54-3 PWT	CN1,CN3	CONN-SMD 3P-P2.54-XH 2.54-WT		ZX-XH2.54-3 PWT	Megastar(兆星)	C7429672	LCSC
11	1	BAT54C	D1	SOT-23-3_L2 .9-W1.3-P1.9 0-LS2.4-BR		BAT54C		C990001352 0	LCSC
12	1	SS34	D2	SMC_L6.9-W 5.9-LS7.9-R D		SS34	onsemi(安森美)	C154551	LCSC
13	1	SS34	D3	SMA_L4.2-W 2.7-LS5.0-R D		SS34	JSMSEMI(杰盛微)	C2837271	LCSC
14	1	SMD1812-11 0-33	F1	F1812		SMD1812-11 0-33	宏嘉诚	C46641155	LCSC
15	1	PZ254V-12-6 P	H2	HDR-TH_6P- P2.54-V-M-R 2-C3-S2.54		PZ254V-12-6 P	XFCN(兴飞)	C492420	LCSC
16	2	2.54-1*4P母	H3,H4	HDR-TH_4P- P2.54-V-F		2.54-1*4P母	BOOMELE(博穆精密)	C2718488	LCSC
17	2	PZ254V-12-1 6P	H5,H6	HDR-TH_16 P-P2.54-V-M -R2-C8-S2.5 4		PZ254V-12-1 6P	XFCN(兴飞)	C492425	LCSC
18	2	PZ254V-12-4 P	H7,H8	HDR-TH_4P- P2.54-V-M-R 2-C2-S2.54		PZ254V-12-4 P	XFCN(兴飞)	C492419	LCSC
19	1	PZ254V-12-8 P	H9	HDR-TH_8P- P2.54-V-M-R 2-C4-S2.54		PZ254V-12-8 P	XFCN(兴飞)	C492421	LCSC
20	1	HDR-M_2.54 _1x4P	H10	HDR-TH_4P- P2.54-V-M				C492403	

21	1	PM254V-11-04-H85	H11	HDR-TH_4P-P2.54-V-F		PM254V-11-04-H85	XFCN(兴飞)	C541851	LCSC
22	1	15uH	L1	IND-SMD_L7.0-W6.6_FXL0630	15uH	FXL0630-150-M	cjiang(长江微电)	C177243	LCSC
23	3	MHT170UGCT	LED1,LED2,LED3	LED0805-R-RD		MHT170UGCT	MEIHUA(美华)	C397047	LCSC
24	1	WJ500V-5.08-2P	P1	CONN-TH_2P-P5.00_WJ500V-5.08-2P		WJ500V-5.08-2P	KANGNEX(康奈克斯电气)	C8465	LCSC
25	1	2.54-1*6P母	P2	HDR-TH_6P-P2.54-V-F		2.54-1*6P母	BOOMELE(博穆精密)	C40877	LCSC
26	2	200kΩ	R3,R24	R0805	200kΩ	0805W8F2003T5E	UNI-ROYAL(厚声)	C17539	LCSC
27	2	40.2kΩ	R6,R26	R0805	40.2kΩ	FRC0805F4022TS	FOJAN(富捷)	C2933438	LCSC
28	1	210kΩ	R7	R0805	210kΩ	FRC0805F2103TS	FOJAN(富捷)	C2933367	LCSC
29	1	47.5kΩ	R8	R0805	47.5kΩ	RCA0547K5FLF	HKR(香港电阻)	C22369969	LCSC
30	1	5kΩ	R9	R0805	5kΩ	ARG05DTC5001	Viking(光颀)	C2828783	LCSC
31	2	5.1kΩ	R10,R11	R0805	5.1kΩ	0805W8F5101T5E	UNI-ROYAL(厚声)	C27834	LCSC
32	1	10kΩ	R12	R1206	10kΩ	1206W4F1002T5E	UNI-ROYAL(厚声)	C17902	LCSC
33	1	2kΩ	R15	R0805	2kΩ	0805W8F2001T5E	UNI-ROYAL(厚声)	C17604	LCSC
34	3	10K	R17,R18,R19	R0603	10K				
35	4	M3螺丝	SCREW1,SCREW2,SCREW3,SCREW4	M3螺丝					
36	1	HX KCD11-2P-R	SW1	SW-TH_2P-L15.0-W10.5-P5.00_KCD11-2P-R		HX KCD11-2P-R	hanxia(韩下)	C6917318	LCSC
37	1	SS-12D10L5	SW2	SW-TH_SS-12D10L5		SS-12D10L5	XKB Connectivity(中国星坤)	C319012	LCSC
38	1	Key_SMD_6x6x6.5	SW3	SW-SMD_4P-L6.0-W6.0-P4.50-LS9.0-2					
39	1	MP4560DN-LF-Z	U1	SOIC-8_L4.9-W3.9-P1.27-LS6.0-BL-EP3.1-1		MP4560DN-LF-Z	MPS(芯源)	C40133	LCSC
40	1	XT60PW-F	U2	TH_2P-XT60PW-F		XT60PW-F	AMASS(艾迈斯)	C428722	LCSC

41	1	LM317MDT	U3	TO-252-3_L6 .5-W6.1-P4.5 8-LS10.0-TL		LM317MDT	UMW(友台半 导体)	C3015165	LCSC
42	1	XT30PW-F20 .G.Y	U4	CONN-TH_X T30PW-F-1		XT30PW-F20 .G.Y	AMASS(艾迈 斯)	C2913282	LCSC
43	1	STM32F103 VET6	U7	LQFP-100_L 14.0-W14.0- P0.50-LS16. 0-BL		STM32F103 VET6	ST(意法半导 体)	C8308	LCSC
44	2	ZX-XH2.54-4 PWT	U8,U13	CONN-SMD 4P-P2.50-W AFER-254W- 4P		ZX-XH2.54-4 PWT	Megastar(兆 星)	C7429673	LCSC
45	1	HC-PM254-8 .5H-1x6PW	U9	HDR-TH_HC -PM254-8.5H -1X6PW		HC-PM254-8 .5H-1x6PW	HC(虹成电子)	C22436147	LCSC
46	2	ZX-XH2.54-6 PWT	U10,U12	CONN-SMD ZX-XH2.54-6 PWT		ZX-XH2.54-6 PWT	Megastar(兆 星)	C7429675	LCSC
47	1	TYPE-C 6P	USB1	TYPE-C-SM D_TYPE-C-6 P		TYPE-C 6P	SHOU HAN(首韩)	C456012	LCSC
48	1	8MHz	X1	CRYSTAL-S MD_4P-L3.2- W2.5-BL_SIT 8008BI	8MHz	XL2EL89CSI -111YLC-8M	YXC(扬兴晶 振)	C5444548	LCSC
49	1	32.768kHz	X2	OSC-SMD_L 3.2-W1.5	32.768kHz	SF32WK327 68D31T002	TKD(泰晶)		LCSC





在接入高压输入的情况下，
严禁接入XT30和USB的5V输入接口

IO扩展

四路继电器1

PA1_PWM1

PA0_PWM2

四路超声波

四路继电器3

串口扩展

HC-05蓝牙接收

IO扩展

E1RS接收机

Figure 1 illustrates the experimental design and the results of the study. The figure is divided into two main sections: **A** (Experimental Design) and **B** (Results).

Section A: Experimental Design

The design is a 2x2x2 factorial design, with the following factors and levels:

- Factor 1: Treatment**
 - Control (C)
 - Low Dose (L)
 - High Dose (H)
- Factor 2: Time Point**
 - Baseline (B)
 - Post-Treatment (P)
- Factor 3: Subject**
 - Subject 1 (S1)
 - Subject 2 (S2)
 - Subject 3 (S3)
 - Subject 4 (S4)
 - Subject 5 (S5)
 - Subject 6 (S6)
 - Subject 7 (S7)
 - Subject 8 (S8)
 - Subject 9 (S9)
 - Subject 10 (S10)

The design is represented by a grid of boxes, where each box contains the treatment, time point, and subject number. The grid is organized into two main groups, **A** and **B**, which are further divided into subgroups. The results of the study are presented in a table below.

Section B: Results

The results are presented in a table with the following columns: **Subject**, **Treatment**, **Time Point**, **Mean Value**, **Standard Deviation**, **Significance**, and **Notes**.

Subject	Treatment	Time Point	Mean Value	Standard Deviation	Significance	Notes
S1	C	B	1.2	0.5		
S1	L	B	1.5	0.6		
S1	H	B	1.8	0.7		
S1	C	P	1.1	0.4		
S1	L	P	1.4	0.5		
S1	H	P	1.7	0.6		
S2	C	B	1.3	0.6		
S2	L	B	1.6	0.7		
S2	H	B	1.9	0.8		
S2	C	P	1.2	0.5		
S2	L	P	1.5	0.6		
S2	H	P	1.8	0.7		
S3	C	B	1.4	0.7		
S3	L	B	1.7	0.8		
S3	H	B	2.0	0.9		
S3	C	P	1.3	0.6		
S3	L	P	1.6	0.7		
S3	H	P	1.9	0.8		
S4	C	B	1.5	0.8		
S4	L	B	1.8	0.9		
S4	H	B	2.1	1.0		
S4	C	P	1.4	0.7		
S4	L	P	1.7	0.8		
S4	H	P	2.0	0.9		
S5	C	B	1.6	0.9		
S5	L	B	1.9	1.0		
S5	H	B	2.2	1.1		
S5	C	P	1.5	0.8		
S5	L	P	1.8	0.9		
S5	H	P	2.1	1.0		
S6	C	B	1.7	1.0		
S6	L	B	2.0	1.1		
S6	H	B	2.3	1.2		
S6	C	P	1.6	0.9		
S6	L	P	1.9	1.0		
S6	H	P	2.2	1.1		
S7	C	B	1.8	1.1		
S7	L	B	2.1	1.2		
S7	H	B	2.4	1.3		
S7	C	P	1.7	1.0		
S7	L	P	2.0	1.1		
S7	H	P	2.3	1.2		
S8	C	B	1.9	1.2		
S8	L	B	2.2	1.3		
S8	H	B	2.5	1.4		
S8	C	P	1.8	1.1		
S8	L	P	2.1	1.2		
S8	H	P	2.4	1.3		
S9	C	B	2.0	1.3		
S9	L	B	2.3	1.4		
S9	H	B	2.6	1.5		
S9	C	P	1.9	1.2		
S9	L	P	2.2	1.3		
S9	H	P	2.5	1.4		
S10	C	B	2.1	1.4		
S10	L	B	2.4	1.5		
S10	H	B	2.7	1.6		
S10	C	P	2.0	1.3		
S10	L	P	2.3	1.4		
S10	H	P	2.6	1.5		